Prediction Depth of Controlled Cellular Automata

A Lean-Verified Classification for One-Sided Rules

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Abstract. We classify the prediction depth of one-sided cellular automata whose frozen control words gate a stay/hop data update over $\mathbb{F}_2$. The input is the explicit initial window and the output is the predicted data bit at its apex. An executable classifier selects a branch for each fixed rule. A single logspace-uniform, polynomial-size De Morgan family for that rule realizes the selected depth bound at every time. Prediction is in logspace-uniform NC^0 exactly when the hop guard is identically zero, or when it is identically one and the stay guard is identically zero. When the hop guard is nonconstant, or is identically one and the stay guard is nonconstant, depth is $\Theta(\log(T+1))$. For the homogeneous Pascal rule at $T > 0$, the minimum depth is $2p(T)+1$ with De Morgan gates, where $p(T)$ counts the set bits of T. The verified uniform generator uses depth at most $3p(T)$. As a structural comparison, a separate one-path model with transported control has positive-time depth one at equal velocities and exactly $\lceil \log_2(T+1) \rceil$ at unequal velocities. Lean 4 verifies the classification, the connection from dynamics to finite prediction, the Pascal optimum, and the uniform generators for the frozen rules.

Keywords. Cellular automata, circuit depth, logspace uniformity, Lean 4.

1 Introduction

Which verifiable conditions on an evolution rule F and an observation O bound the circuit depth of $O\big(F^{T(x)}\big)$ for a specified input encoding and circuit model? Algebraic structure can make cellular automaton prediction much shallower than layer-by-layer simulation [1]. Here we ask which fixed guards in a one-sided, two-track automaton admit constant-depth prediction and how depth depends on the observation time.

Our main result is the four-way classification of frozen rules (Theorem 2). A finite guard test selects the branch, and one logspace generator for the chosen branch achieves the stated upper bound at every time. The main construction uses the forced order of hops to summarize all paths to an endpoint by two prefix statistics. A binary no-carry test then detects which coefficients survive cancellation (Section 3.5). The homogeneous branch has a nonmonotone depth profile governed by the set bits of the observation time. A separate one-path rule shows how relative velocity changes the number of distinct initial controls read (Theorem 3).

2 Preliminaries

A local rule on configurations over $\mathbb{Z}$ is iterated T times. We ask for one data bit, at site zero. Each model in the main results has a one-sided radius-one update, so its initial window consists of sites $0, \ldots, T$. For the frozen model the input contains $q(T+1)$ control bits followed by $T+1$ data bits. The control width q and the two guard circuits are fixed independently of T. The legal word lengths are $n = (q+1)(T+1)$ and determine T uniquely. The prediction language $L_{s,h}$ consists of the legal words whose predicted bit is one, with all other lengths rejected.

Our upper bounds use Boolean AND and OR gates of fan-in two, NOT gates, and constants. An input wire has depth zero and a NOT gate costs one layer. Gates can share outputs and connect arbitrary sites. The lower bounds in the classification apply to any Boolean gates of fan-in at most two. For a fixed rule, write $D(T)$ for the minimum De Morgan circuit depth on the legal window at time T and write $\ell(m) = \lceil \log_2 m \rceil$ for $m \geq 1$. Uniform circuit families are indexed by input length. A deterministic

logspace transducer receives the length in unary and writes the complete circuit code, including its gates, wiring, and output. We use logspace uniformity throughout the statements about NC.

A coordinate is *essential* if two assignments that differ only at that coordinate give different outputs.

Essential-input bound. A depth-d circuit of fan-in at most b has at most b^d essential coordinates. For $b \geq 2$, a function with $m \geq 1$ essential coordinates therefore requires depth at least $\lceil \log_b m \rceil$.

Proof. Only input wires in the backwards cone of the output can affect it. At depth zero the cone has at most one input. At the next depth a gate has at most b predecessor cones, so induction bounds the number of input wires by b^d. In a shared circuit, counting a repeated predecessor more than once only increases the bound. □

3 Frozen-control automata

3.1 The iterate formula

Let $a_i \in \{0,1\}^q$ be a control word, $x_i \in \mathbb{F}_2$ an initial data bit, and $s, h : \{0,1\}^q \to \{0,1\}$ the stay and hop guards. The synchronous update is

$$a_i' = a_i, \quad x_i' = s(a_i)x_i + h(a_i)x_{i+1}, \tag{1}$$

where the data arithmetic is in $\mathbb{F}_2$. A backwards path ending at site k has k hops and $T-k$ stays. Put

$$e_k = \prod_{j=0}^{k-1} h(a_j), \quad w_k = \sum_{j=0}^{k} s(a_j), \tag{2}$$

where the second sum is in the natural numbers. For natural w, r, write $M(w,r)$ for the number of weak compositions of r into w parts. For $w > 0$, it is $\binom{w+r-1}{r}$, and $M(0,r)$ is one at $r = 0$ and zero otherwise.

Theorem 1 (Iterate formula). *For every initial configuration and time T, the data bit at site zero is*

$$x_0(T) = \sum_{k=0}^{T} e_k (M(w_k, T-k) \bmod 2) x_k. \tag{3}$$

The same identity holds at any site after translating the indices.

Proof. Expanding the local recurrence once per step expresses the output as a sum over backwards paths. Each hop must be enabled, contributing the factor e_k. A path visits the sites 0 through k in order. The number of stays allocated to each enabled site determines the path, including possible stays after the last hop. There are $M(w_k, T-k)$ such allocations. Their contributions add in $\mathbb{F}_2$, giving Equation 3. The Lean proof establishes the corresponding coefficient recurrence by induction on the iterate of Equation 1. The formula involves only the initial window. □

3.2 Depth classification

Write $p(T)$ for the number of ones in the binary expansion of T, and d_s, d_h for the depths of the fixed guard circuits. A guard varies if it takes both Boolean values. All the bounds below are finite, including $T = 0$.

Guard condition	Lower bound	Uniform upper bound
$h \equiv 0$	0	$1 + d_s$
$h \equiv 1$, $s \equiv 0$	0	0
h varies, or $h \equiv 1$ and s varies	$\ell(T+1)$	$O(\log(T+2))$
$s \equiv h \equiv 1$	$p(T)$	$3p(T)$

Table 1: Depth bounds at legal lengths for fixed frozen guards.

Theorem 2 (Uniform classification). *For each fixed pair of guard circuits there is one logspace-uniform, polynomial-size De Morgan family that decides the whole language and achieves the appropriate upper bound in Table 1 at every legal length. It has depth $O(\log(n+2))$ at all lengths. The lower column bounds $D(T)$ and holds even for arbitrary fan-in-two Boolean gates. Constant-depth membership is characterized by*

$$L_{s,h} \in \text{logspace-uniform } \mathrm{NC}^0 \quad \Leftrightarrow \quad h \equiv 0 \vee (h \equiv 1 \wedge s \equiv 0). \tag{4}$$

3.3 Proof of the uniform classification

Local and shift cases. For $h \equiv 0$ and $T > 0$, the answer is $s(a_0)x_0$ by idempotence. The pure shift case returns x_T. At $T = 0$ every rule returns x_0.

Varying guards. When h varies, set all data bits except x_T to zero. The surviving term in Equation 3 is $x_T \prod_{j=0}^{T-1} h(a_j)$, regardless of s. A nonconstant Boolean guard has an essential bit, as seen along a single-bit path between two words with different guard values. Each of the first T control words contributes one essential bit on this restriction, and x_T contributes one more. When $h \equiv 1$ and s varies, fix $s(a_0) = 1$ and $s(a_i) = 0$ at sites 1 through T. Then $w_k = 1$ for every k and Equation 3 is parity of all $T + 1$ data bits. Both cases give the same lower bound by essential coordinates.

The upper bound in this branch is given by the construction in Section 3.5.

Homogeneous Pascal rule. For $s \equiv h \equiv 1$, the coefficients are $\binom{T}{k} \bmod 2$. Over $\mathbb{F}_2$, the identity

$$(1+z)^T = \prod_{j:\mathrm{bit}_{j(T)}=1} \left(1 + z^{2^j}\right) \tag{5}$$

shows that precisely the binary submasks of T occur. Their $2^{p(T)}$ data bits are essential. The logarithmic lower bound is therefore $p(T)$. A balanced XOR tree reduces those leaves in $p(T)$ rounds. Each XOR can be implemented with at most three De Morgan layers. The uniform generator scans $k = 0, \ldots, T$, emits the submask wires, and serializes a regular XOR reducer with $5p(T)2^{p(T)}$ gates.

A single uniform family. The guard circuits and the selected branch are fixed. The logspace transducer writes that branch's predictor at legal lengths and a rejecting circuit at other lengths. Its polynomial-length output bounds the number of gates. The local and shift branches have a constant bound. In the varying branch the lower bound grows with T. For the homogeneous branch it grows along the times $T = 2^r - 1$. This proves the constant-depth characterization. □

The accompanying classifier tests constancy of the guards by traversing control words by prefix. Lean proves that its return value is equivalent to the branch condition. A constancy test stops at its first counterexample and can visit 2^q words in the worst case.

3.4 Exact depth in the Pascal case

Write $p = p(T)$ and $N = 2^p$. The Pascal output is parity of N inputs. With XOR available as a single binary gate, a balanced tree attains the lower bound p. With De Morgan gates the exact depth is zero at $T = 0$ and $2p + 1$ at $T > 0$.

Proof. For the upper bound, give every leaf both its bit and its negation. A node with child parities a, b computes $a \oplus b$ and its complement in two layers from $a, \neg a, b, \neg b$. For example, the positive output is $(a \wedge \neg b) \vee (\neg a \wedge b)$ and the negative output is $(a \wedge b) \vee (\neg a \wedge \neg b)$. The p internal levels therefore add $2p$ layers to the one layer of leaf negations.

For the lower bound, set irrelevant inputs to constants and unfold the circuit into a formula of the same or smaller depth. Push negations to the leaves and simplify constants. Khrapchenko's parity bound requires at least N^2 literal leaves [2]. Pushing negations preserves leaf count and simplifying constants can only decrease it. A formula of depth at most $2p$ has at most $2^{2p} = N^2$ leaves, with equality only for

a full binary tree with all leaves at depth $2p$. That tree has only AND and OR gates and hence computes a monotone function. Parity of $N \geq 2$ inputs is nonmonotone. Thus depth at most $2p$ is impossible. □

Lean verifies both per-time optima. The De Morgan profile drops from $2r + 1$ at $T = 2^r - 1$ to three at $T = 2^r$, for $r \geq 2$. The uniform classification uses the separate $3p$ generator in Table 1.

3.5 Uniform prediction circuit

Let $B(a)$ be the set of one-bit positions in the binary expansion of a. For $w > 0$, Lucas's theorem says that $M(w, r)$ is odd exactly when $B(r) \subseteq B(w + r - 1)$ [3]. Equivalently, adding r and $w - 1$ produces no carry. Indeed, if the bits of r are contained in those of the sum, subtracting r clears those bits without a borrow. The remaining bits are those of $w - 1$. Including the case $w = 0$, the coefficient test is

$$M(w, r) \bmod 2 = 1 \quad \Leftrightarrow \quad (w = 0 \wedge r = 0) \vee (w > 0 \wedge B(r) \cap B(w - 1) = \varnothing). \tag{6}$$

The hop prefix e_k decides whether a path can reach k. Once the hops are enabled, the stay count w_k suffices because the enabled sites can each absorb any number of stays. For example, with two enabled stay sites and one stay to allocate, both possible paths cancel. In Equation 6 this is the shared low bit of $r = 1$ and $w_k - 1 = 1$.

From the test to a circuit. Evaluate the guards in parallel. Compute each e_k by a balanced AND tree and each w_k by a carry-save counter. Each merge round compresses two pairs of words into one pair at constant depth, and the final addition propagates carries across only $O(\log(T + 2))$ bits. For each k the generator knows $r = T - k$. It checks Equation 6 for each candidate count $j = 0, \ldots, k + 1$ by a binary scan, and the circuit tests equality to w_k for the candidates that pass. At most one count matches. Gate the selected bit by $e_k x_k$, then take parity over the sites. This uses a fixed number of logarithmic-depth stages and polynomial size, even with a separate counter for every prefix.

The verified generator. Lean verifies a shared decoder that can handle any time u within the window. It generates the table for all such u using the Lucas bit test, and selects the rows for T with a one-hot tag. The tag implements the composition interface by routing wires according to the binary digits of T. Each parity bank reads the low bit of a carry-save counter. A logspace transducer emits the complete gate and output lists of this circuit, using recomputation to resolve references. The explicit depth bound is $104(\ell(T + 3) + 1) + \max(d_s, d_h)$. Appendix A describes the layout and the resource accounting.

4 Transported controls as a structural comparison

In this separate model the control track has its own dynamics and the data follows one path. Fix velocities $u, v \in \{0, 1\}$ and a Boolean control and data track (c_i, d_i). Define the synchronous rule

$$c_i' = c_{i+u}, \quad d_i' = c_i \wedge d_{i+v}. \tag{7}$$

The window has $2(T + 1)$ bits. With a single backwards data path, the data at site i and time T equals

$$d_{i+vT} \wedge \bigwedge_{k=0}^{T-1} c_{i+kv+(T-1-k)u}, \tag{8}$$

where the right-hand side uses initial bits and the conjunction is empty at $T = 0$.

Theorem 3 (Exact depth under transport). *Depth is zero at $T = 0$. For $T > 0$, it is exactly one if $u = v$ and exactly $\ell(T + 1)$ if $u \neq v$. The formal upper-bound construction uses a padded AND tree with at most $2m$ binary AND gates for m essential inputs. The lower bound applies to all fan-in-two Boolean circuits predicting the data bit.*

Proof. At the k-th step of the backwards path we read control at site $i + kv$ and time $T - 1 - k$. The initial site of that control bit is $i + kv + (T - 1 - k)u$, which proves Equation 8. At $T = 0$ the answer is the input wire d_i. At positive times with equal velocities all control indices coincide, leaving a conjunction of one control bit and one data bit. In general, the initial control index is $i + (T - 1)u + k(v - u)$.

For unequal velocities the indices are $i, \dots, i+T-1$ in forward or reverse order. Every distinct factor is essential, since flipping it with all other factors set to one changes the output. Thus there are two essential coordinates in the equal case and $T+1$ in the unequal case. For m factors, a balanced AND tree of height $\ell(m)$ attains the essential-input lower bound. Padding to the next power of two uses $2^{\ell(m)} - 1 \le 2m$ binary gates. □

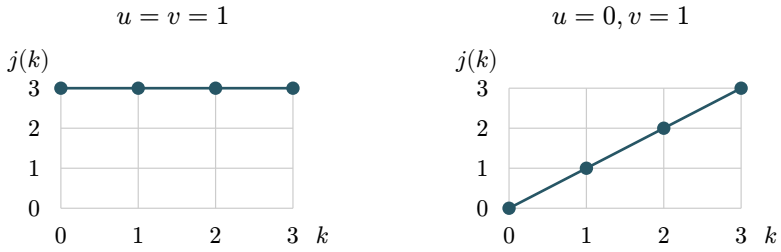


Figure 1: Initial control site $j(k)$ at backward step k for $i = 0$ and $T = 4$. Both paths end at the initial data bit d_4.

Figure 1 keeps the data path fixed and varies only u. The pair $(u, v) = (0, 0)$ is the frozen no-hop case with stay guard c_i. Changing only u to one gives logarithmic depth although the data never hops. For $(u, v) = (1, 1)$ the data and control both move, yet their equal velocity gives constant depth. For this single-path rule, the relevant count is the number of distinct initial controls encountered in the moving frame of the data path. The time-dependent framework in [1] accommodates this rule. For finite Boolean monomial systems, the link between iterate support and fixed-length dependency walks is classical [4]. On the finite prediction window, the two shifts make that support explicit.

5 Evolving controls with branching

Uniform transport. Allow both terms in Equation 1 and transport the entire control track by $a_{i(t+1)} = a_{i+1}(t)$. In the reflected moving frame $y_{j(t)} = x_{-j-t}(t)$, substitution gives

$$y_{j(t+1)} = h\big(a_{-j-1}(0)\big)y_{j(t)} + s\big(a_{-j-1}(0)\big)y_{j+1}(t). \tag{9}$$

The guards are now frozen and exchanged. The observation $x_0(T)$ is $y_{-T}(T)$, whose data window is the reversal of the original one. The frozen control window is $a_{T-1}(0), \dots, a_0(0)$ followed by a padding word. At the last virtual site no stay can occur, so the padding is irrelevant. The two prefix statistics still work in this frame, with the roles of stay and hop exchanged. Lean verifies the coordinate change and the per-time circuit reduction, including the input rewiring.

Additive control evolution. Now let the control be one bit and evolve it by $c_{i(t+1)} = c_{i(t)} + c_{i+1}(t)$, with data update $x_{i(t+1)} = c_{i(t)}x_{i(t)} + x_{i+1}(t)$. Both updates use $\mathbb{F}_2$. At $T = 2$, the coefficient of $x_1(0)$ at the origin is $(c_0(0) + c_1(0)) + c_1(0) = c_0(0)$. The initial pairs $(1, 0)$ and $(0, 1)$ have the same stay count and hop flag for this endpoint, but yield different coefficients. A stay can now read different guard values at the same site. Counting how many sites allow a stay at time zero loses the order information needed for cancellation.

This rule still has a short coefficient description. Put $Q_{T(i)} = c_{i(T)} + c_{i+T}(0)$, the Pascal parity of the initial controls with its rightmost term removed. For $0 \le k < T$, put

$$P_{T,k}(i) = \prod_{j=k+1}^{T-1} \big(1 + c_{i+j}(0)\big). \tag{10}$$

Then the coefficient of $x_{i+k}(0)$ in $x_{i(T)}$ is $P_{T,k}(i)Q_{T(i)}$ for $k < T$, and one for $k = T$. Thus a single parity is shared across endpoints, and a suffix-zero test selects which endpoints receive it.

Proof. Write $b_{T,k}(i)$ for the coefficient and take it to be zero outside the window. The update gives

$$b_{T+1,k}(i) = c_{i(T)} b_{T,k}(i) + b_{T,k-1}(i+1). \tag{11}$$

Since a bit squares to itself, $c_{i(T)} Q_{T(i)} = (1 + c_{i+T}(0)) c_{i(T)}$. For $k < T$, the stay term therefore has suffix factor $P_{T+1,k}(i)$, as does the shifted term because $P_{T,k-1}(i+1) = P_{T+1,k}(i)$ for $k > 0$. For $k = 0$ the shifted term is absent, but $Q_{T(i+1)}$ already vanishes whenever $P_{T+1,0}(i) = 1$: the suffix then zeros every initial control in that parity. After factoring out the suffix the sum is $c_{i(T)} + Q_{T(i+1)} = Q_{T+1}(i)$. For $k = T$ the stay term is $c_{i(T)}$ and the shifted term is $Q_{T(i+1)}$, giving the same sum with empty suffix. This also holds at $T = 0$ since $Q_0(i+1) = 0$. The all-hop term is always one. □

A block of consecutive sites has summary (Z, V), where Z tests whether its controls are all zero and V is the parity of its data bits with only zero controls to their right within the block. A single site has summary $\big(1 + c_{j(0)}, x_{j(0)}\big)$. Concatenating blocks combines their summaries by

$$(Z_L, V_L), (Z_R, V_R) \mapsto (Z_L Z_R, V_L Z_R + V_R). \tag{12}$$

A balanced tree computes V on the first T sites, with empty summary $(1, 0)$. The prediction is $x_{T(0)} + Q_{T(0)} V$. A second tree computes $Q_{T(0)}$ on the submask wires. The two trees use $O(T+1)$ gates in total and have depth $O(\log(T+2))$. Their regular layouts can be streamed in logspace.

At positive times the essential inputs are exactly the first T control bits and all $T+1$ data bits. Setting all but $x_0(0)$ to zero on the data track gives $x_0(0) c_0(0) \prod_{j=1}^{T-1} \big(1 + c_{j(0)}\big)$, which witnesses the control bits. Setting $c_0(0) = 1$ and $c_1(0) = \ldots = c_{T-1}(0) = 0$ gives parity of the data bits. This yields the depth lower bound $\ell(2T+1)$. Lean verifies the coefficient formula for the actual coupled iterate. The circuit bounds follow from the argument above.

6 Related work

Pascal parity and fast prediction. The odd-entry count $2^{p(T)}$ is classical. Rowland attributes the binary count to Glaisher and derives it from Lucas's theorem [3]. Moore uses Pascal coefficients to predict homogeneous additive rules, obtaining logarithmic depth with binary gates and logspace-uniform ACC^0 with unbounded modular sums [1]. Robison gives a serial method for computing an individual coefficient from one initial seed [5]. The per-time optimum in Section 3.4 follows from classical parity complexity. The verified uniform family realizes the Pascal profile within a constant factor.

Dependent linear tracks and the coefficient formula. In [1], Lemma 1 composes spacetime transfer coefficients by dividing the time interval. Theorem 1 applies that algorithm to quasidirect products. Examples 1 and 3 cover binary data tracks controlled by another track and triangular linear tracks such as our additive-control example. With balanced XOR trees for the modular sums, the general method gives an $O\big(\log^2(T+2)\big)$ bound for our binary tracks. Time invariance and the one-sided stencil reduce the frozen coefficients to prefix statistics. In §3, Salo studies the sitewise choice between identity and right-neighbor XOR on a finite ring [6]. That example uses the local maps of $s \equiv 1$, $h(a) = a$ and concerns surjectivity and periodicity. Here the control word is part of the input and the output is one finite-window prediction bit.

Parallel prefix circuits are classical [7]. Balanced products, carry-save counting, and coefficient tests give the frozen predictor's $O(\log(T+2))$ bound. The coefficient in Equation 3 also has a classical matrix analogue. The one-step data map is upper bidiagonal on the relevant finite window. Higham's Theorem 6.1 for functions of such matrices gives the product of intervening off-diagonal entries times a divided difference in the diagonal entries [8]. For the monomial z^T this divided difference is a complete homogeneous polynomial of degree $T - k$. On zero-one diagonal entries it counts the same weak compositions. The polynomial identity can be specialized to integer zero-one entries before reducing modulo two, with no division in $\mathbb{F}_2$.

Mechanization and the meaning of frozen. Silváši and Tomášek formalized bounded grids and computable cellular automata in Lean [9]. Their work supplies precedent for machine-checked cellular automata. The present artifact checks a rule-family depth classification and the actual circuit generator. The freezing model of Ollinger and Theyssier imposes a monotone order on each whole cell state [10]. In Equation 1 only the control word is fixed, and the data at a site can cycle. The contribution here is the executable four-way classification, the matching logarithmic bounds for varying guards, and the proof that one logspace generator per rule achieves the stated upper bounds across all lengths. The comparisons above leave priority for this combination open.

7 Mechanization and scope

The window interface separates infinite dynamics from finite inputs. `rawSample` reads the window from a configuration, and `decodeRaw` extends an arbitrary legal window to an infinite configuration. Lean proves that sampling this extension recovers the supplied bits. Correctness on sampled configurations therefore gives correctness on every legal input. Lifting from time to input length selects that same predictor at legal lengths and a rejecting circuit at the others.

The uniformity proof generates the code of the circuit used for the depth bound, including the variable-length lists of gates and output references. The reusable code combinators verify reference bounds and relocation for parallel banks and serial composition. These obligations allow a generator to recover individual references by recomputation in logarithmic space. The classification theorem uses the selected circuit family as one witness for correctness, size, uniformity and the per-time bound. The transport theorem establishes optimal depth separately at each window length.

Each frozen control word selects one of four local data maps 0, x_i, x_{i+1} and $x_i + x_{i+1}$. Once the controls are fixed, the data dynamics is one-sided and linear. The input is an explicit window with length linear in T, and the circuit can wire across arbitrary sites.

The accompanying Lean development formalizes three sufficient criteria for prediction problems. A compiles resource-bounded composition circuits into a predictor. B transfers P-hardness from circuit evaluation given a logspace simulation witness. Exclusion from uniform NC via B assumes $\mathrm{P} \neq \mathrm{NC}$ and is a complexity-class statement. C gives an unconditional depth lower bound from essential inputs. At fixed fan-in, dimension, radius, and bit width, a polynomial-volume light cone limits the **counting bound** from C to $O(\log(T + 2))$. Actual prediction depth can exceed this bound. Lean checks supplied witnesses. Finding witnesses for arbitrary rules is a separate problem. In the frozen family, the executable classifier selects the depth branch by testing guard constancy on a fixed finite alphabet.

8 Conclusion

For frozen controls, guard constancy decides the constant-depth cases, and prefix statistics give a uniform logarithmic-depth predictor for varying guards. The Pascal depth is governed by the set bits of the observation time. With branching, uniform transport preserves the frozen summary after a change of frame, while additive control evolution needs a different summary even with all hops enabled. In the latter example, the induction closes because the all-hop coefficient is fixed and the control evolves by Pascal convolution. It remains to determine when a comparable coefficient recurrence closes if the evolving control also gates the hops.

Artifact availability

The Lean and Typst sources and the build instructions are in the project repository. Use the repository snapshot accompanying this PDF for the declarations listed in Appendix B. The repository documents the pinned Typst and font inputs for rebuilding all four editions. Run `./scripts/check.sh` for the strict Lean build, transitive axiom audit, and negative regression. The repository's verification guide separates build evidence from review of the model assumptions. Appendix B lists the declarations behind the main results.

A Generator layout and depth budget

Put $m = T + 1$ and $B = \ell(T + 3) + 1$. The register stores the time tag τ, the stay and hop flags, and the data in four blocks of width m. The tag starts at index zero. Reading the binary digits of T from most significant to least, each stage doubles the index and increments it when the digit is one. These operations route wires with zero fill.

At site k, the decoder enumerates $0 \le u < m$ and $0 \le j \le k + 1$. For a row with $u \ge k$ and odd $M(j, u - k)$ it emits the test $(w_k = j) \wedge \tau_u$. All other rows are zero. Parity over j, then u, selects the coefficient. The decoder multiplies by the hop prefix and data and takes parity over k.

The normalized counter tree costs at most $2 + 14b$ layers, including initialization and copying in each merge round, for a counter of word width $b = \ell(a + 1)$ on a bits. The finishing addition costs at most $1 + 8b$. Together they cost at most $22(b + 1)$. The counts therefore cost at most $22B$, and taking the low bit of a counter for parity is bounded by $23B$ including the empty case. The depth proof charges $5B$ for a row test, from the literal comparisons, conjunction and time tag. The hop conjunction, bounded by $5B$, runs alongside the lookup. The decoder budget is $22B + 5B + 3 \cdot 23B + 2 = 96B + 2 \le 98B$. Register initialization costs at most two layers with normalization. There are at most $2\ell(T + 1)$ subsequent register stages of at most two layers each, also counting normalization. This gives at most $6B$ for the register path and a total of $104B + \max(d_s, d_h)$ including guards.

The table has $m^2 \frac{m+3}{2}$ rows and there are $O(m^2)$ counters of at most $m + 1$ input bits. The regular layouts give $O(m^3 \log^3(m + 1))$ gates for fixed guards. A gate record is indexed by its stage, block, bit lane and position in a local template. For parallel banks the serializer counts earlier blocks to obtain a wire offset. In a chain of width W, identity gates $y \wedge y$ give the stage outputs consecutive indices. After G gates on N primary inputs, output slot a of that stage is wire $N + G - W + a$. This lets the serializer recompute references with counters of logarithmic width and fixed nesting depth. It streams the gate and output lists from the unary length.

B Lean correspondence

The paths below are relative to `PredictionDepth/`.

Result	Lean module and declaration
Frozen iterate	`FrozenPascal/Semantics.lean`
	`iterate_data`
Window interface	`FrozenPascal/GuardRules.lean`
	`rawSample_decodeRaw`
Executable classifier	`FrozenPascal/Classification.lean`
	`classify`, `classify_eq_iff`
Uniform classification	`FrozenPascal/UniformClassification.lean`
	`uniform_depth_classification`
	`predictionLanguage_mem_uniformNC_zero_iff`
Uniform predictor	`FrozenPascal/UniformPrediction.lean`
	`uniformGuardedPrediction_code_logspace`
	`uniformGuardedPrediction_depth`
Code composition	`Complexity/CodeLogspace.lean`
	`compCode_logspace`
Pascal support and XOR depth	`FrozenPascal/HomogeneousPascal.lean`
	`pascalSupport_card`
	`homogeneousNativeXorPrediction_depth`
Exact Pascal De Morgan depth	`FrozenPascal/PascalExactDepth.lean`
	`homogeneous_prediction_deMorgan_exact_depth`
Transported control	`TransportedControl.lean`
	`iterate_data_true_iff`, `essential_prediction_iff`
	`exact_prediction_depth`, `constant_depth_iff`
Branching under transport	`TransportedBranching.lean`
	`reverseFrame_iterate`, `prediction_correct`
	`prediction_depth`, `prediction_size`
Additive-control coefficients	`AdditiveBranching.lean`
	`trimmedParity_eq_sum`, `iterate_data`

References

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